

Pastry Pirates v3

The Redesign

UI/UX audit, second pass — blue-sky, researched against the best digital board games

Pass one read as tweaks — fair. This pass starts from the other end: what would v3 look like built the way the best digital adaptations are built, and what is the staged path from here to there inside your constraints (vanilla JS, one file, deterministic engine untouched)?

The north star, in one sentence

The board is the app. One full-screen stage with a camera — pinch, pan, and a director that frames each moment — everything else riding on top of it: a ribbon of game state above, a thumb-zone sheet of decisions below, and every object on the water inspectable with a tap.

Every mock in this document is the real v3 build, restyled live and screenshotted at 375×812 — including the zoom shots, which are the game's actual camera (an SVG viewBox change), not an image blow-up. The hero shot on page 3 is your own board art doing what it has always been capable of.

Numbers correction from pass one: the grid is 15×15, not 13 — sail squares are ~**21px** at 375px width, worse than reported. At the proposed default zoom they become ~45px: over your 44px bar. Zooming is not a nice-to-have; it is how every number in this game gets fixed at once.

What the best adaptations do

Five patterns recur across the digital board games people actually praise — and each maps straight onto Pastry Pirates.

1 · The board is a canvas with a camera

Ticket to Ride, Root, Terraforming Mars, Scythe — every serious adaptation ships pinch-zoom and pan as table stakes, with helpful indicators for valid placements. The underlying pattern is the "zoom UI" of Maps and Figma: a world larger than the screen, explored through a camera. v3 instead scales its whole world down to 319px and asks a thumb to hit a 21-pixel square.

2 · Actions live where the thumb lives

Mobile ergonomics research is unambiguous: primary actions belong in the bottom 40–50% of the screen; top corners are the dead zone. The bottom-sheet pattern — content sliding up over a full-bleed canvas — is the one interaction model designed around where the thumb actually is. v3 currently scatters decisions down a scrolling page, with state (your recipe) below the fold.

3 · Everything is inspectable

In the praised adaptations, tapping any object answers "what is this and what does it mean for me?" — Through the Ages will even explain why an action is *unavailable*. v3's board is pure display: islands, rival ships, the rim current — none of them answer a tap.

4 · Teach by doing, in character

Through the Ages' tutorial — "Ancient Vlaada" walking you through your first civilization — is widely called the best in digital board gaming, and the app won BGG's Golden Geek Best App on the strength of it. Root ships one scripted tutorial per faction. Nobody praised ships a rules modal as their only teacher. v3 has exactly one teacher, and it is a 5.3-screen rules modal that has drifted from the engine.

5 · Ceremony for the moments of chance

Wingspan's reputation was built on atmosphere — animation and sound doing UX work, making digital feel *more* alive than cardboard, not less. Dice towers, card flips and coin tosses get theater in good adaptations because chance is the emotional spike of a board game. v3's flippenator — the single most consequential object in the game — is a 54px button that flips in place while a clock counts down beside it.

The stage — all of it together

[BIG] · north star · principles 1 + 2

Full-bleed board filling the phone. A ribbon above: round, turn order as captain ships (the active one ringed), wind now, weather to-morrow. A sheet below, in the thumb zone: the narration voice, the decision buttons, and yer recipe as a live tracker. No page scroll — the board is the app, and the camera has already framed your turn.



THE TARGET — the live build, restaged: viewBox camera + ribbon + thumb-zone sheet

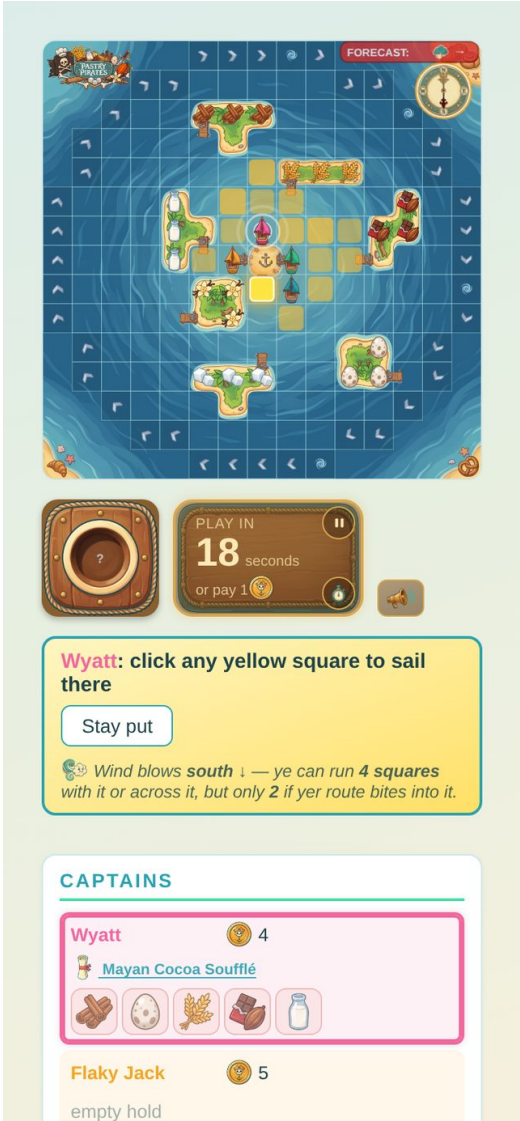
Everything in this shot is your existing art rendering at native resolution — the gold sail squares here are ~90px tap targets. The captains panel becomes the four ships in the ribbon (tap = expand a captain card, finding N3); the footer moves behind a menu on the sheet; the flip coin appears in the sheet when — and only when — a flip is wanted.

N1 · The camera

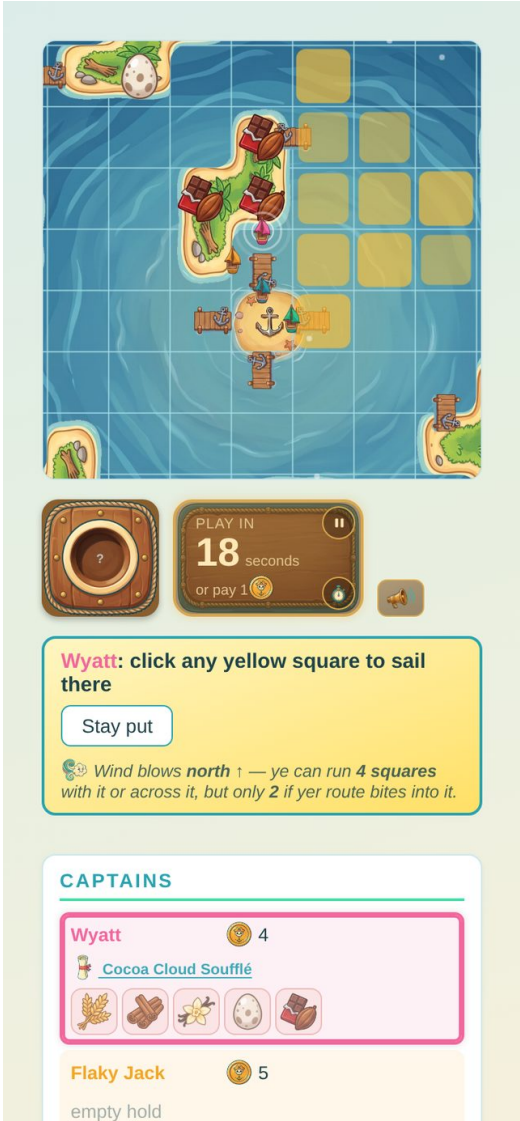
[MEDIUM] · the zooming recommendation · buildable first

Three parts, shippable inside the current layout before any other change:

- **Pinch to zoom, drag to pan, double-tap to toggle** fit-board ↔ 2.1× on your ship. At 2.1× a sail square is ~45px — past your 44px bar with no visual redesign at all.
- **A director.** Your turn: glide to your ship. A battle: push in on the two ships. A storm: pull back to the whole board and let the shove read. Bake-off: pan to Tortuga. Any player gesture instantly overrides the director — the camera is staff, not captor.
- **The mechanism already exists.** The board is one SVG; the camera is its viewBox. The zoom mock below is literally a one-line viewBox change on the live game — full asset resolution, no blur. Animating it is ~40 lines of vanilla JS (lerp the four numbers under rAF); pinch/pan is a pointer-event handler. Guests and replay are untouched: the camera is pure render-side state.



BEFORE — whole world at 319px, 21px cells



AFTER — the same SVG, viewBox-zoomed: ~45px cells

N1b · The sailing view: the whole window, and to-morrow's wind

[MEDIUM] · your two rules, built in

Two constraints you set now govern the sail turn, and both are in the mock below, live:

- **The camera never crops a legal move.** On a sail prompt the director frames the bounding box of every reachable square plus your ship, with a squares-wide margin — and only then zooms as far as that window allows (capped at 2.2×). This shot settled at 1.32× because the wind gave a wide window: all 17 reachable squares are on screen, guaranteed by construction. A tight window zooms closer; a sprawling one pulls back. Pinch and pan still override.
 - **The forecast is a first-class instrument.** A two-card wind HUD sits in the quiet corner of the frame: THIS TURN with a large arrow, NEXT TURN beneath it — reading straight from the engine (windNow / forecastWind()). When a storm is coming the NEXT card turns storm-red: "■ STORM — heading hidden," keeping your design decision honest instead of contradicting it. The HUD supersedes the 7.5px on-board chip during play; the compass dial stays as board art.
- The capture got lucky in the best way: wind north this turn, south next — the exact reversal a player must see coming to not waste this turn's sail. That is the case this HUD exists for.





DESKTOP — same two rules, HUD beside the board

N2 · The stage and the sheet

[BIG] · kills page-scrolling · principle 2

The layout stops being a document and becomes a stage:

- **Board:** full-bleed behind everything (preserveAspectRatio: slice), camera from N1.
- **Ribbon (top):** round number · the four captains as their ships in turn order, active one ringed — tap any ship for their card · wind now · forecast. The 7.5px chip problem dissolves: the forecast lives in a real UI layer at UI scale, while the compass stays on the water as art.
- **Sheet (bottom):** three states. *Peek* — one line of narration + recipe tracker. *Half* — the current decision with its buttons (every prompt the game has today renders here unchanged; the narration machinery and reveal order move in whole). *Full* — captains detail, log, settings. Drag or tap between states; the game never asks for a page scroll again.
- **What it costs:** this is the one genuinely structural build — a repot of existing organs (panel, captains, footer) into three fixed containers. The engine, the prompts, the narration scheduler, multiplayer lockstep: untouched. It is CSS + a sheet controller, not a rewrite.

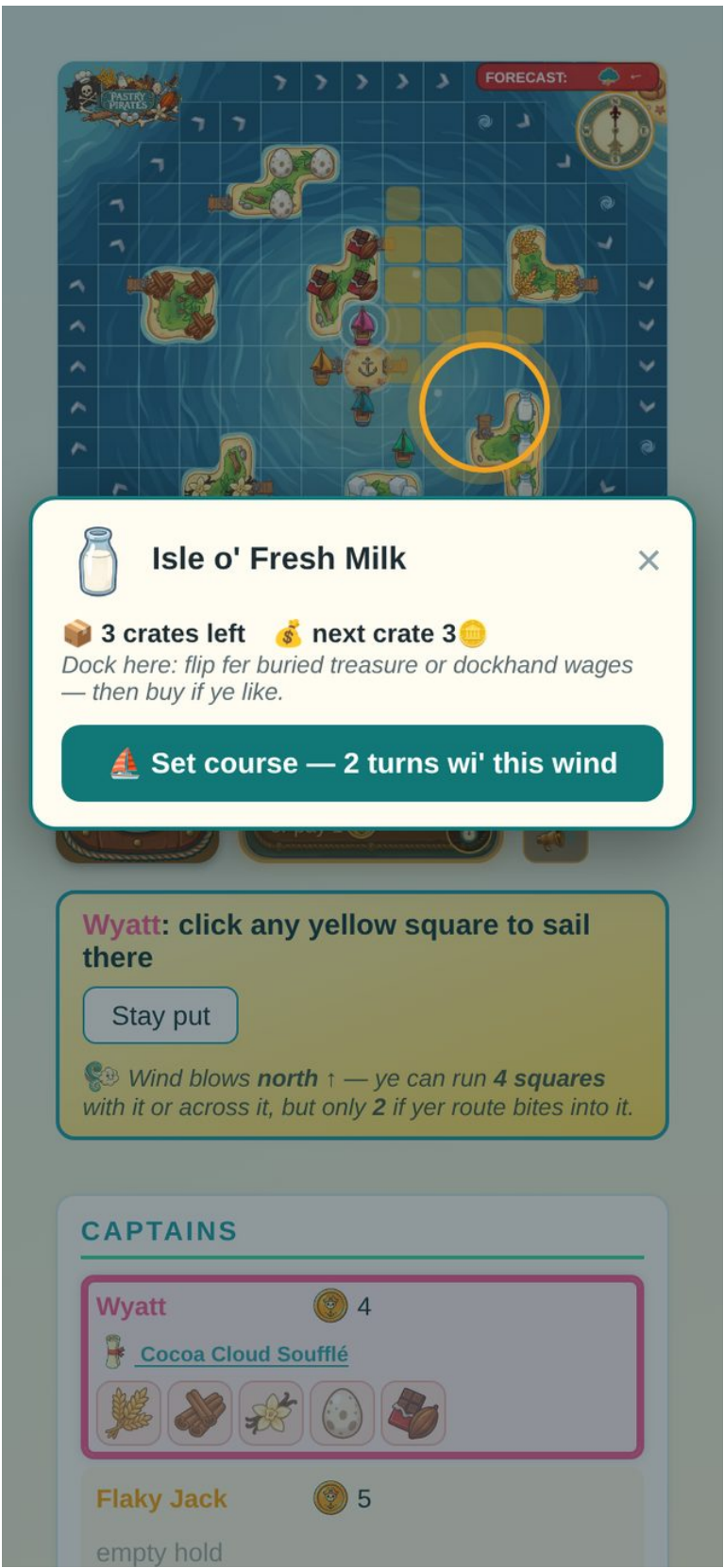
The hero shot on page 3 is this recommendation, live. A note on what it retires: the mute button placement question, the captains-below-the-fold problem, the footer's seven-button pile, and the fold itself — the whole class of "where does X fit at 375px" findings from pass one stop being questions.

N3 · Tap anything, learn something

[MEDIUM] · principle 3 · teaches the economy passively

Every object on the water answers a tap with a card:

- **An island:** its ingredient, crates left, next-crate price — live from the engine (the mock below shows real state: 3 crates, next one 3 coins) — plus the dock rule in one line, and "Set course" with a wind-true ETA (N5).
- **A rival ship:** captain, coins, hold, and — in range? — the attack option with its price. The "can't attack an empty hold" rule appears exactly when you're looking at an empty hold.
- **The rim:** "the trade winds — free passage round the board, once ye're in the current." The least discoverable mechanic in the game becomes one curious tap away.
- **Tortuga:** the ovens, who's home, what firing them up takes.



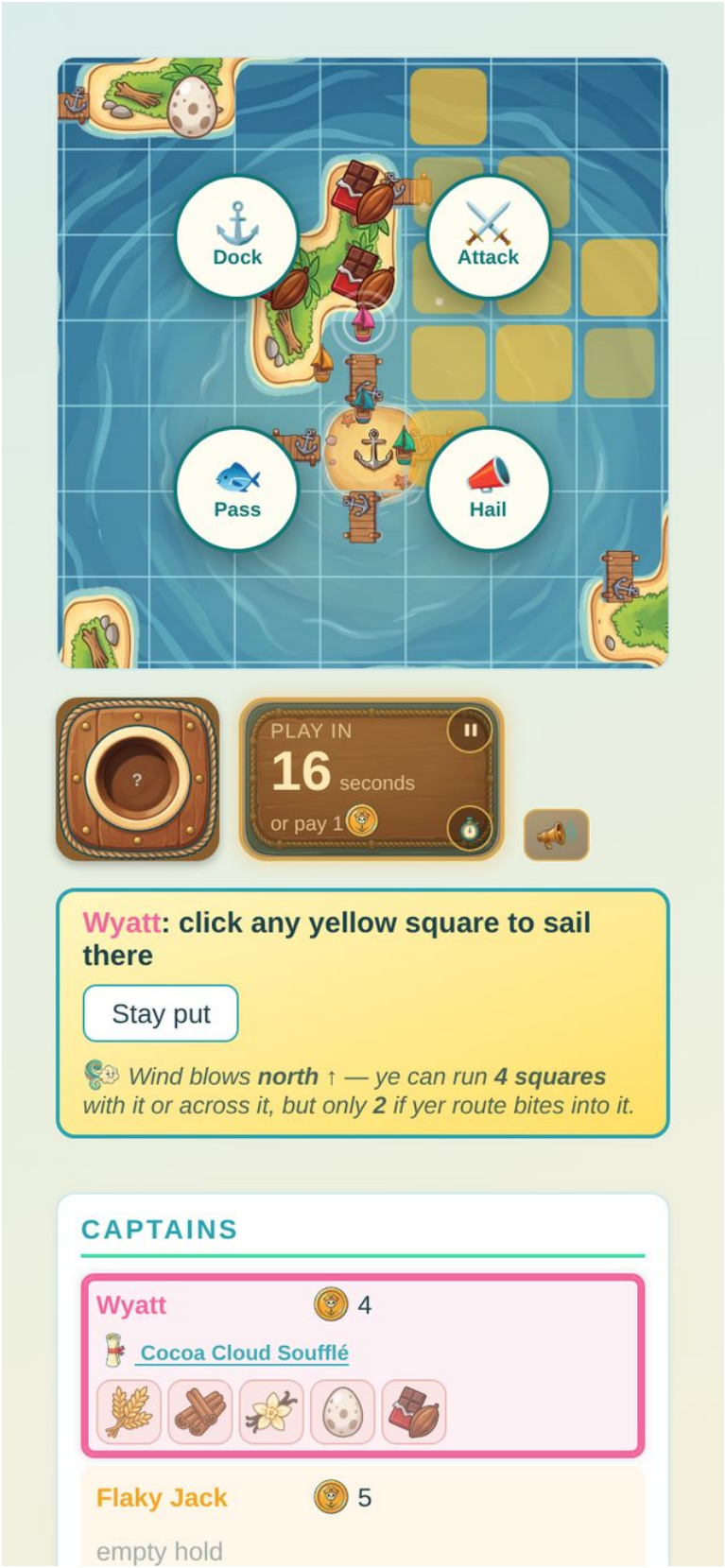
LIVE MOCK — real engine data: 3 crates left, next 3 coins

This is Through the Ages' deepest lesson: the interface answering "why / what / how much" at the point of curiosity replaces a rulebook chapter. Nothing here invents state — every number already exists in the engine (tokens, cratePrice, holds, positions).

N4 · Actions at the ship

[MEDIUM] · pairs with the camera

After you sail, the choices bloom around your ship — Dock, Attack, Hail, Pass — right where your eyes already are, instead of in a panel a scroll away. With the camera zoomed (N1) the targets are enormous; disabled options render greyed with their reason attached, your D-41 pattern moved to the site of the action. On the un-zoomed board the same four buttons dock into the sheet — the radial is an enhancement, not a dependency.



LIVE MOCK — the after-sail choice, at the ship

N5 · Sail by intent — tap the goal, see the voyage

[MEDIUM→BIG] · teaches wind math by showing

Tap a crate in your recipe tracker (or an island card's "Set course") and the board answers: a dashed course from your bow to that dock, with a wind-true ETA — "≈ 2 turns wi' this wind." The upwind cap stops being a paragraph anyone has to read; it is visible as the difference between a 2-turn course and a 4-turn one when the wind swings.

- **Medium:** preview only — the route draws, the ETA shows, you still tap squares to sail. The engine's own reachability already computes everything the ETA needs.
- **Big:** commit from the route — "Sail it" takes the first leg now and remembers the heading, one less decision every turn once you've set intent. (The race-planner bot already plans voyages this way; this hands the same instrument to the human.)



LIVE MOCK — course + wind-true ETA to the wheat dock

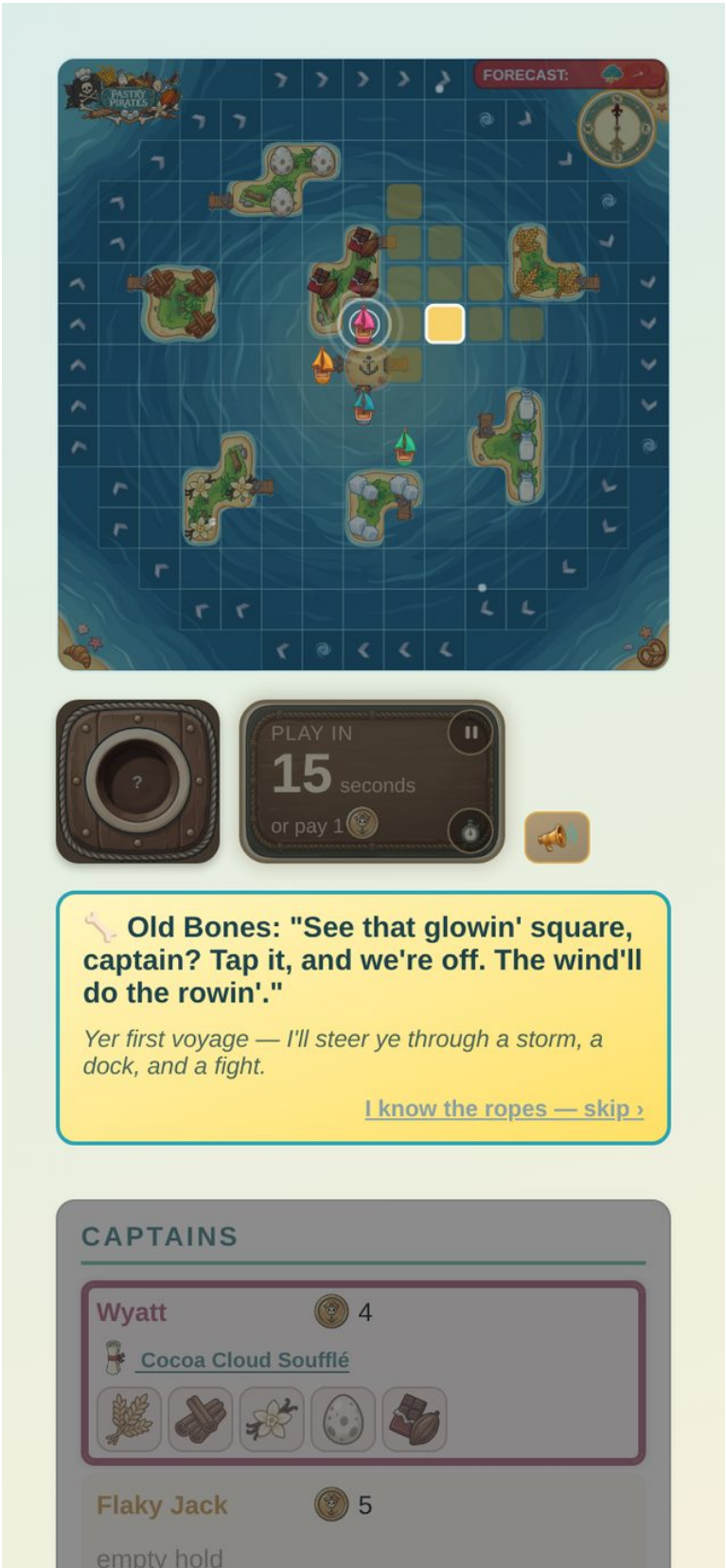
N6 · The guided first voyage

[BIG] · principle 4 · the deferred tutorial, designed

Not a rules modal, not hint toasts — a scripted first voyage in character, the Through the Ages model. A narrator — **Old Bones**, say, a retired captain — sails your first three turns with you on a fixed seed chosen so the script always works:

- Turn 1: one square glows. "Tap it, and we're off — the wind'll do the rowin'."
- Turn 2: the wind swings against you; your reach visibly shrinks to 2. "See how she fights ye? Never beat upwind if ye can run across it."
- Turn 3: dock, flip (the coin gets its ceremony, N7), buy a crate as prices are explained by the island card (N3). A bot demonstrates the rim current on cue. Storm on round 3, scripted, shove watched from the pulled-back camera. Then: "Yer on yer own, captain" — and the real voyage continues from exactly that state.

"I know the ropes — skip" sits on screen at every beat; the whole script is one seed + ~10 gated prompts on top of the existing localAsk flow. Every rule taught by doing, inside 90 seconds, zero pages of prose.



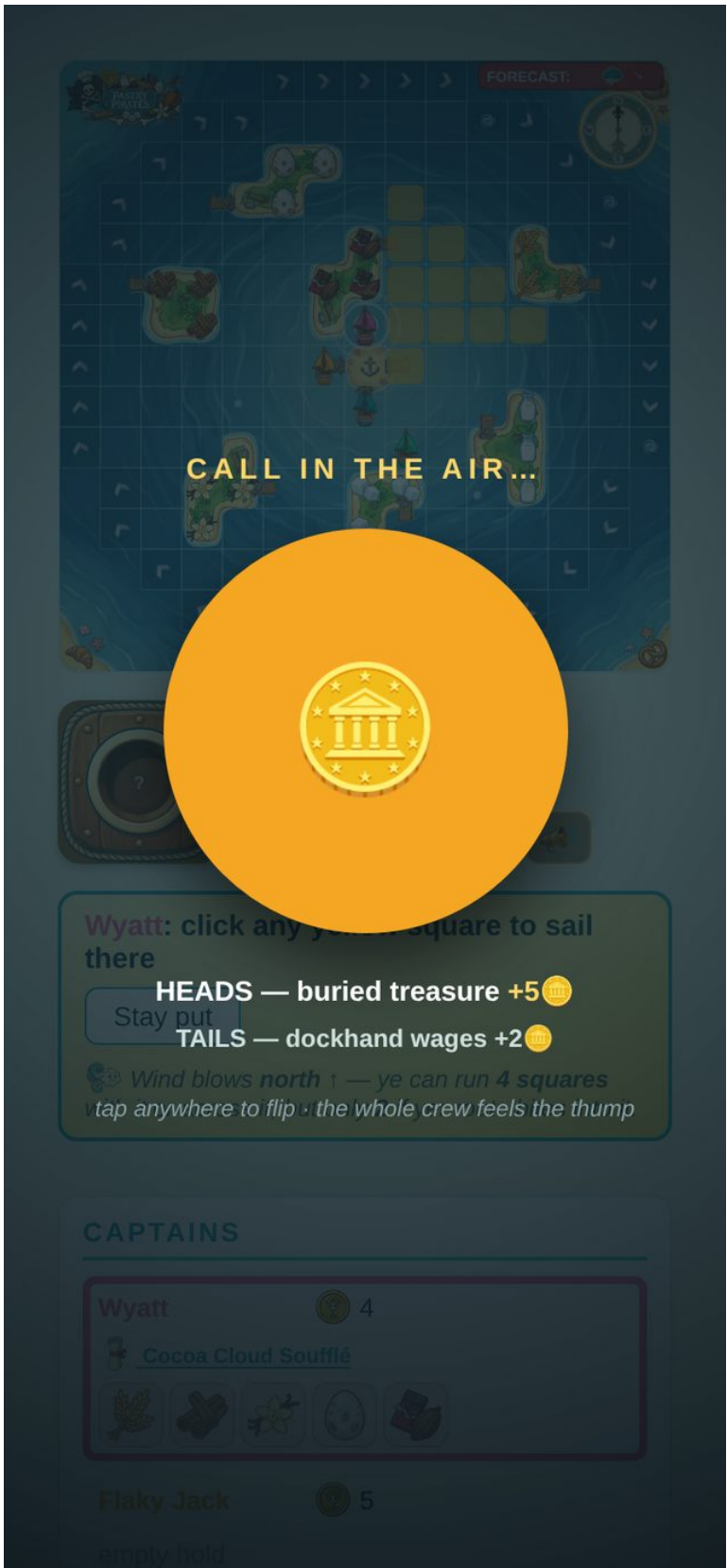
LIVE MOCK — one glowing square, one narrator line, a visible skip

N7 · Chance deserves ceremony

[MEDIUM] · principle 5 · the flippenator's promotion

When a flip decides something, the world holds its breath: board dims, the coin takes the screen, stakes printed under it — heads and what it pays, tails and what it pays — one tap, a real spin, a haptic thump on landing, and the result stamps itself before the world returns. Two seconds of theater, once or twice a turn, and the game's central ritual finally feels like one. (Wingspan's whole reputation is this principle applied relentlessly.)

Same treatment, scaled up, for the broadside volley — both coins in the air at once — and the bake-off reveal. The clock (if any) pauses during ceremony; pass one's finding that the turn clock ticks through battles gets fixed by design here, not by patch.



LIVE MOCK — the flip as a held-breath moment, stakes visible

N8 · The feel layer

[MEDIUM] · spec, no mock — this one has to be felt

A one-week pass that makes every mechanism legible through motion and touch, extending what your wind dots and storm rain already do:

- **Haptics map** (iOS Safari 18.4+ / Android vibrate): light tick on sail commit, thump on coin landing, double-thump on broadside, long-low on storm arrival, success pattern on badge.
- **Camera punch:** 4px kick on a broadside hit; slow 1.02× breathe on your-turn start; the storm's pull-back IS the storm cue.
- **Wake trails:** ships leave a fading wake as they move — bot turns become watchable theater instead of teleports, and the storm shove draws its own explanation.
- **Sound:** your Phase-21 taste (layering, table-wide hearing, storm fading out) already reads like a feel spec — the missing wind/cannon/new-round/your-turn cues slot here.

What pass one becomes

Every pass-one finding survives inside this architecture — most stop being patches and become properties: guided first minute → N6's opening beat · sail tap size and mis-taps → N1 camera · illegible instruments → N2 ribbon · teach-in-play → N3 cards + N6 script · recipe visibility → the sheet's tracker · solo clock → clockless practice voyage in N6 · how-to-play drift → fix the three facts now regardless (still page-one work) · EoV chrome, 44px sweep, copy fixes → still worth shipping this week while the stages land. The full pass-one report sits beside this one in the repo (docs/ux-audit-v3/, pass-1 file).

N9 · Narration on the water — no box at all

[BIG] · your request · the variant where the board tells the story

The narration box disappears. Every line renders where its story happens:

- **Lines about a captain attach to their ship** — a speech bubble tailed to the hull. Your turn call, a dock outcome, a battle taunt: each floats over the ship that owns it, holds while it is the live line, then drifts up and fades when the next line replaces it. The game already has this machinery — the sea-creature chat bubbles (chatBubbles in board.js) anchor to ships and reposition on every render. This variant promotes that system from decoration to the narrator itself.
- **Lines about the table hover over the water** — round headers, wind swings, storm calls appear as a banner over the open sea, then drift and fade.
- **Decisions keep a thumb home.** Buttons cannot float mid-ocean — they stay in a slim peek strip (mobile) or float beside the ship (desktop), but the WORDS live on the board.
- **The log keeps every word.** Bubbles are ephemeral by design; the captain's log becomes the persistent record, one tap away.



YOUR TURN — the call attached to yer ship



A BOT'S DOCK — the outcome at its ship



TABLE-WIDE — round header hovering over the water

What carries over from your narration rulings: the reveal discipline holds (one live line, fade-then-replace, never two competing) — it just happens in space instead of in a box. What to watch: two simultaneous stories (a battle while a trade resolves) need a queue so bubbles never overlap; spectators in multiplayer read the same bubbles, so the “never empty” rule becomes “the active ship always has the line.” Worth a real playtest against the sheet variant (N2) — they can also blend: bubbles for events, the sheet only when a decision needs buttons.

The desktop stage

[BIG] · the same architecture, wide

Same three layers, recomposed for a 1440px window: the board finally at full glory — 812px, 54px cells, every instrument legible — ribbon on top, and the sheet becomes a right-hand rail. Hover exists on desktop, so inspection gets a peek-on-hover, pin-on-click pattern, and the radial menu is optional (actions can live in the rail).



DESKTOP HERO — board at 812px, decision card + captains in the rail

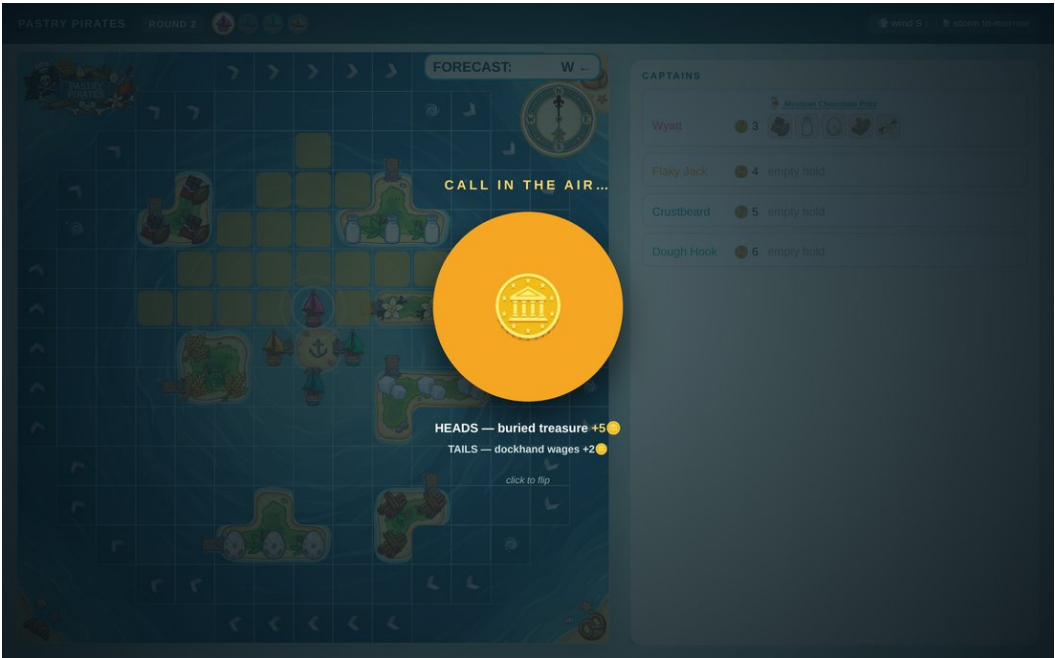


DESKTOP + N9 — bubbles on the water, slim rail, decision floating at the ship

Desktop flow details



INSPECT + ROUTE — hover to peek, click to pin; course + ETA on the big board



THE FLIP CEREMONY — same held-breath moment, desktop scale

And the first decision of the game, wide: the recipe cards sit side by side natively at this width — the compare problem simply does not exist on desktop — and the island glow from N3/N9 teaches the map before the first sail.



RECIPE CHOICE — native side-by-side, ingredient isles glowing for the focused card

Implementation note: one stylesheet, two compositions. The stage/ribbon/sheet-or-rail split is a CSS grid decision at a single breakpoint (~900px); the camera, bubbles, cards and ceremony are identical code on both. Nothing about the engine or lockstep differs by width.

The path from here

Four stages, each shippable alone, each leaving the game better if you stop there. All of it is render-side vanilla JS in the single file; the deterministic engine and multiplayer lockstep are never touched.

Stage A — the camera (N1). `viewBox` lerp + pinch/pan/double-tap + the director's three moves. Also this week: the three drifted rules-modal facts, the 44px hit-area sweep, EoV chrome. Small, loud win — the game feels native immediately.

Stage B — the stage (N2). Full-bleed board, ribbon, three-state sheet; captains and footer repotted; page scroll retired. The one structural build — everything after it gets cheaper.

Stage C — the conversational board (N3 + N4 + N5-medium). Island/ship/rim cards off live engine state, radial actions, route preview + ETA.

Stage D — the soul (N6 + N7 + N8). Old Bones' first voyage, the flip ceremony, the haptic/feel pass. This is the stage a cold player remembers.

Sources

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- Mobile game UI/UX technical guidance: appnality.com/blog/guide-to-mobile-game-ui-ux-design

Method: all mocks are the live v3 build restyled in-page and screenshotted at 375×812 (headless Chromium, dpr 2); zoom shots are real `viewBox` renders. Engine numbers (crates, prices, grid size) read from live state. Corrected this pass: grid is 15×15 → ~21px cells at 375px.